

MH370 flaperon drift diagnostic - native-grid HYCOM

Uncalibrated family-specific diagnostic; currents=hycom_gofs31_reanalysis | isotope=enabled as a neutral-or-penalizing screen

Estimate at a glance

CURRENT FAMILY

hycom_gofs31_reanalysis

PEAK SOURCE CELL

arc-160

PEAK POSITION

-34.463 deg , 93.671 deg

WEIGHTED TRAJECTORIES

2048 ? 77 cells

MEAN IMPORTANCE-WEIGHTED FIELD-TERMINATION FRACTION

72.6%

MEAN PROPOSAL TERMINATION FRACTION

57.3%

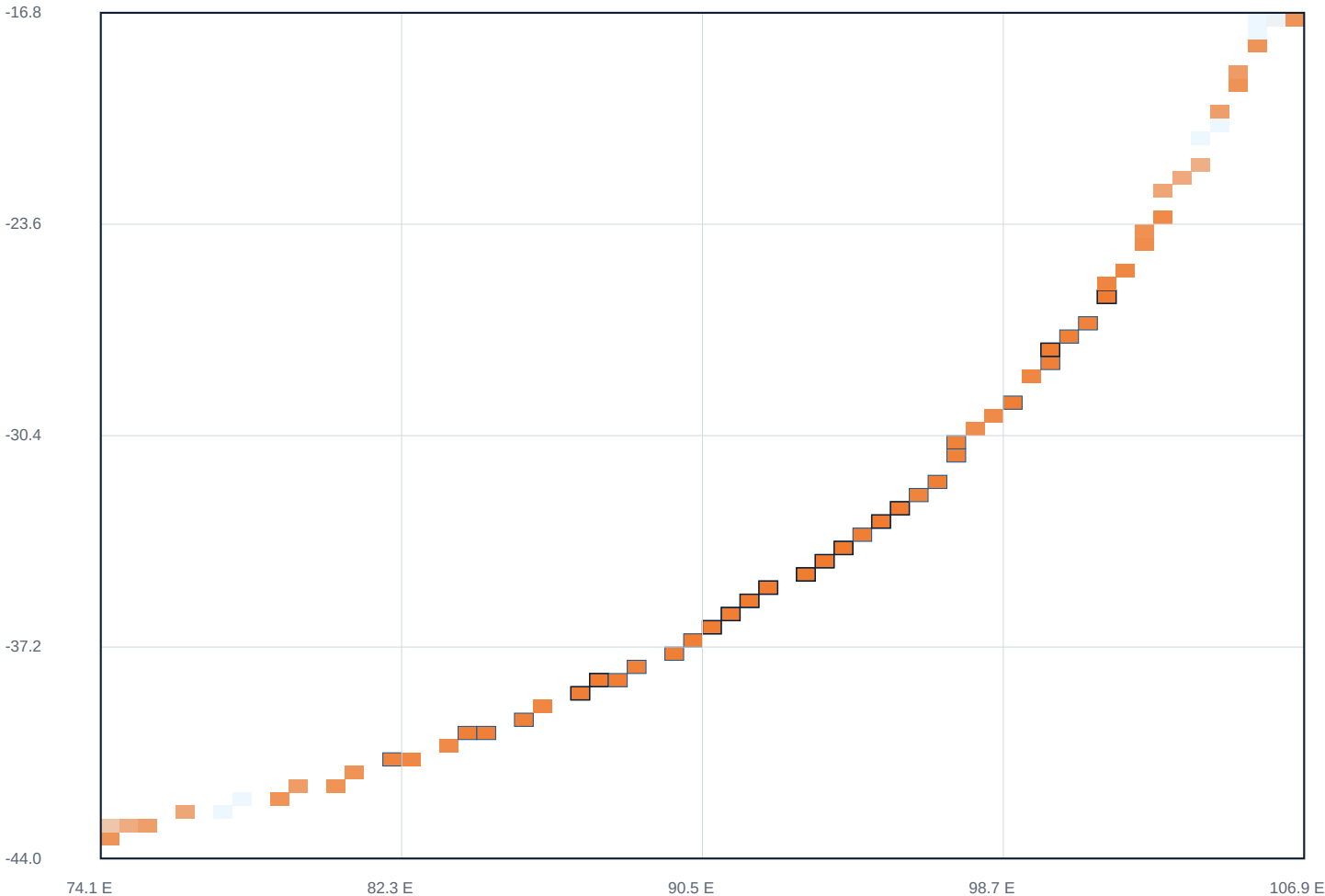
ARRIVAL SAMPLER

AdaptiveSmc

PEAK CORRECTION ESS

847.6

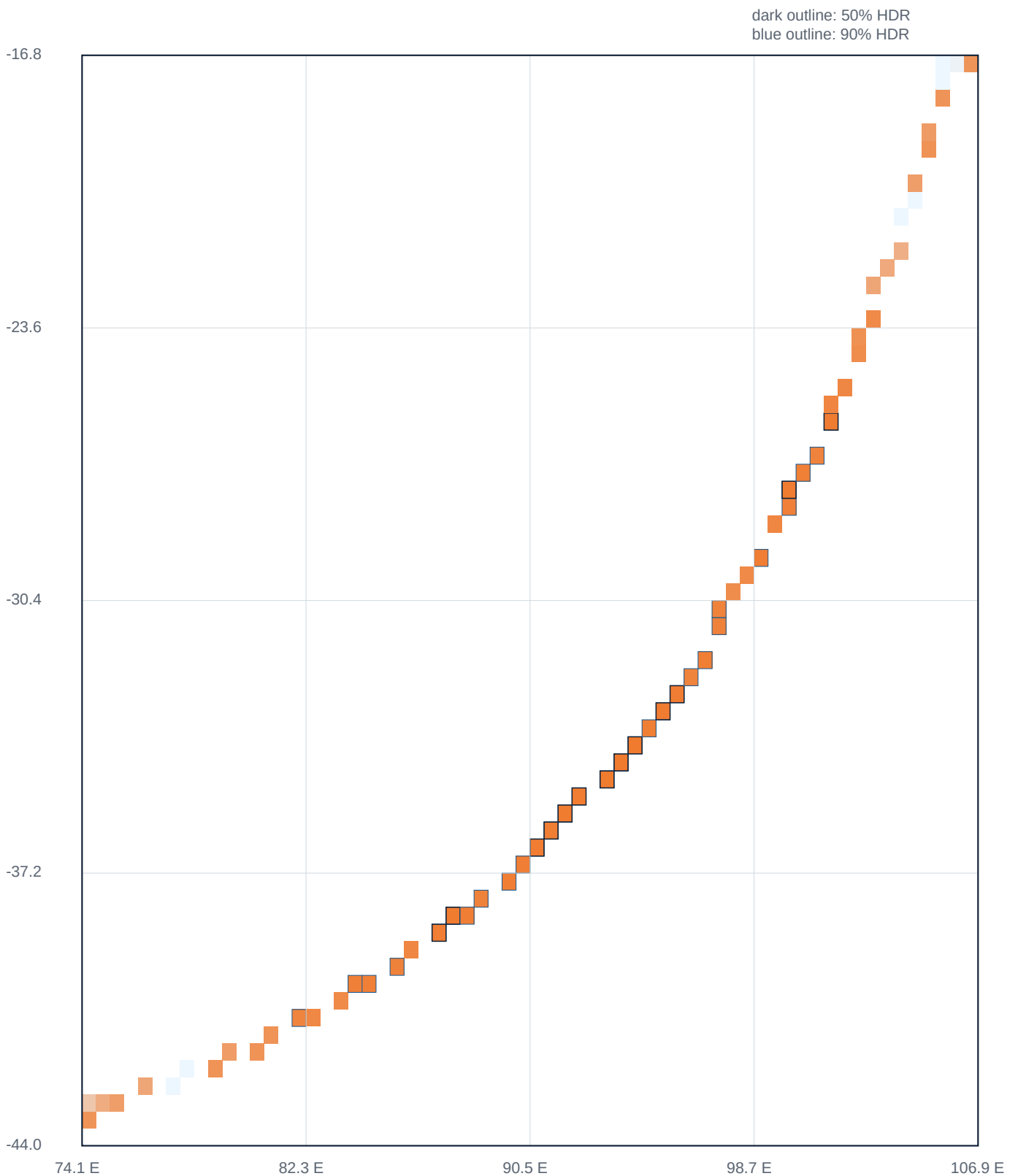
dark outline: 50% HDR
blue outline: 90% HDR



Probability is conditional on the named motion, SATCOM, and evidence configuration.

Posterior geography

Highest-density regions are computed from the weighted particle population.

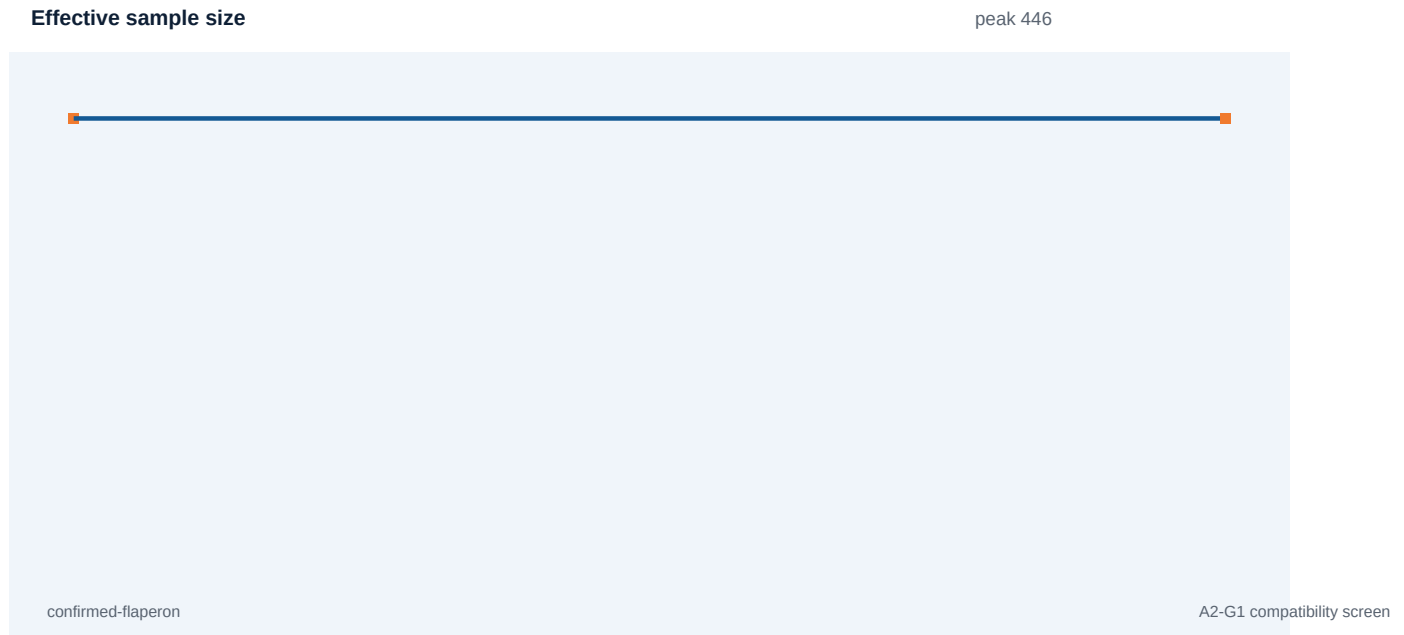


Interpretation

The map is a probability density over the modelled impact state. It is not a searched-area elimination map, and structural model alternatives must be compared separately.

Numerical behaviour and conditions

Adaptive likelihood tempering, particle diversity, and declared evidence conditions.



Evidence conditions

- Current family selected explicitly: hycom_gofs31_reanalysis
- Barnacle isotope: enabled as a neutral-or-penalizing screen conditional on represented flaperon arrival
- direct windage disabled
- distinct Stokes velocity field enabled
- explicit land fraction and calibrated beaching field enabled
- Current interpolation across timestamp gaps greater than 36.0 h is outside-time support
- compatible=1363 marginal=25 rejected=0 missing_SST=550
- Isotope sensitivities: seawater SD=0.20?, calibration SD=1.00 deg C, residual SD=1.50 deg C correlated over 14.0 d, SST SD=0.50 deg C, shell smoothing=7.0 d, compatible/rejection simultaneous confidence=0.950/0.995, maximum marginal/rejection penalties=0.693/4.605 log units

Material limitations

- Diagnostic source-compatibility result, not a calibrated final MH370 posterior.
- Current families are reported separately and are never averaged using subjective weights.
- A2-G1 covers only the final months; the screen is neutral for any compatible chronology and never density-ranks compatible paths.
- No field-unavailable trajectory is relabelled as beaching; termination fractions and reasons remain in the JSON artifact.
- This is every eighth available 3-hour analysis, not a daily mean; exact supplemental analyses repair two stride-created gaps, leaving variable 24-30 hour spacing and phase changes.
- The GOFS 3.1-like reanalysis documents known deep-water and Philippine Sea issues; predictive drifter validation is required before primary use.
- Land remains the packed missing sentinel. No missing ocean value is converted to zero current.
- Stokes drift, direct windage, and coastal beaching are not included in this current field and require distinct explicit inputs.